MULTI-ORFT: Stable Online Reinforcement Fine-Tuning on Diffusion-Based Multi-Agent Planning for Cooperative Driving

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***Abstract*—Closed-loop cooperative driving requires generating multimodal and interaction-realistic multi-agent trajectories while optimizing safety and traffic efficiency. Although diffusion planners pretrained by imitation can capture multimodal interaction realism, they remain weakly aligned with closed-loop cooperative objectives, and *stable online* reinforcement learning (RL) post-training in reactive multi-agent settings remains challenging. We propose MULTI-ORFT, a scene-consistent and cooperative multi-agent planner that *explicitly couples* scene condition-enhanced diffusion pre-training with stable online RL post-training. For pre-training, we build a multi-agent denoising network through inter-agent self-attention and dual-path scene conditioning to enhance scene consistency and road adherence of joint trajectories. For post-training, we develop a stable online RL framework that co-designs dense, well-shaped rewards and variance-gated group relative policy optimization (VG-GRPO), thus strengthening safety- and efficiency-oriented cooperative behaviors under closed-loop execution. Extensive experiments demonstrate that Multi-ORFT delivers superior closed-loop planning performance, achieving consistent gains on core safety and efficiency metrics against state-of-the-art open-source baselines on WOMD and alternative post-training paradigms. The project page is available at: https://anonymous-ee.github.io/multiagentdrrving/.**

***Index Terms*—Reinforcement learning, diffusion policy, closed-loop planning, Multi-agent autonomous driving**

I. INTRODUCTION

Cooperative driving is a *safety- and efficiency-critical* multi-agent planning problem [1], [2]: vehicles must execute right-of-way, yield, merge, and proceed while continuously reacting to one another. Moreover, real interactions are inherently *stochastic* and *multimodal*: multiple safe, rule-compliant interaction patterns may arise in the same scene. As a result, achieving *closed-loop cooperative objectives* while preserving *multimodal*, *interaction-realistic joint* behaviors remains challenging, limiting real-world deployment [3].

Diffusion planners provide a flexible way to model the *joint* distribution of multi-agent trajectories from demonstrations [4], [5]. Their denoising process produces temporally consistent trajectories and supports multimodal sampling, thereby enabling dexterous behaviors with diverse interaction modes. However, existing methods often struggle to balance *multi-agent interaction modeling* and *scene-conditioned modeling*, resulting in joint trajectories that may violate scene constraints [6]–[8]. Furthermore, these methods suffer from *objective misalignment* and *distribution shift* [9]: imitation learning optimizes *data likelihood* instead of explicit human preferences or constraints, making it difficult to optimize safety and efficiency objectives in multi-agent systems. Moreover, closed-loop execution moves the system off the demonstration distribution, where compounding errors and sparse supervision for *rare safety*-critical events increase the risk of out-of-distribution failures. Collectively, these issues limit the design of *safe* and *efficient* diffusion-based multi-agent planning.

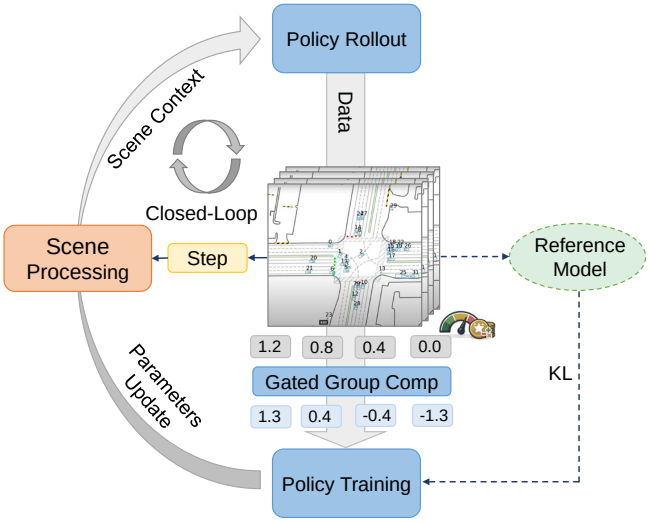


Fig. 1. The proposed stable online RL post-training framework comprises three core components: closed-loop policy rollouts, dense, well-shaped reward evaluation, and variance-gated group relative policy optimization (VG-GRPO).

Reinforcement learning (RL) offers a compelling alternative for enhancing pretrained planning models by coupling sampling-based exploration with reward-driven policy optimization [10], [11]. Recent studies show that RL post-training can enhance planning quality and shape driving patterns, potentially enabling robust closed-loop performance and cooperative behaviors beyond what pre-training alone can achieve [12]–[14]. However, existing work is often reported in largely *offline* settings [15]; without *online* interaction with reactive environments, constrained rollouts and limited exploration often yield only modest closed-loop gains [10], [14]. To go beyond these gains, two ingredients are crucial: (i) *online* interaction, which exposes the policy to scenarios and corrects behaviors under reward guidance, thereby mitigating distribution shift, and (ii) dense and *well-shaped* rewards, which provide fine-grained learning signals for *safety*- and *efficiency*-critical objectives while remaining compatible with realistic, human-like behaviors. Despite these benefits, online RL in *fully* closed-loop settings remains challenging for stringent stability and controllability due to non-stationary coupled interactions, compounding rollout errors, and high-variance gradient estimates.

To address these limitations, we propose MULTI-ORFT, a scene-consistent and cooperative multi-agent planner, which couples scene condition-enhanced diffusion pre-training with stable online RL post-training. In pre-training, we build a multi-agent denoising network on Diffusion Transformers [16] and strengthen scene consistency and constraint adherence of multi-agent trajectories via a dual-path design: cross-attention injects contextual tokens, while AdaLN-Zero provides scene-driven feature modulation to improve road adherence under dense interactions. As shown in Fig. 1, we develop a stable online RL post-training framework to strengthen safety- and efficiency-oriented cooperative behaviors under closed-loop execution. We stabilize online training by co-designing dense, well-shaped rewards and a variance-gated group relative policy optimization (VG-GRPO) strategy: the rewards provide stable, fine-grained supervision aligned with safety and efficiency objectives, while VG-GRPO adaptively gates sampled groups and switches normalization based on within-group reward variance. Our main contributions are summarized as follows:

- We propose a scene condition-enhanced multi-agent trajectory diffusion model that couples cross-attention for direct conditioning injection with AdaLN-Zero for flexible, stable conditional modulation and constraint enforcement throughout the denoising stack.
- We develop a stable online RL post-training pipeline that jointly designs dense, well-shaped rewards and a variance-gated group relative policy optimization (VG-GRPO) strategy to mitigate advantage collapse and gradient instability, thereby significantly improving closed-loop planning performance and cooperative behaviors.
- Extensive experiments demonstrate MULTI-ORFT's superior closed-loop planning performance, particularly on core safety and efficiency metrics. The proposed online framework produces safe, interactive, and efficient cooperative trajectories that outperform baselines on WOMD [17] and other post-training methods, highlighting strong robustness and practical potential.

II. RELATED WORK

A. Multi-agent Behavior Modeling in Traffic Scenarios

Modeling the joint behavior of interacting agents is fundamental to autonomous driving, yet remains challenging because future motion is intrinsically multimodal, long-horizon, and tightly coupled across participants [3]. Earlier forecasting pipelines capture multi-modal behaviors via parametric continuous distributions (*e.g.*, Laplace [18]) or learnable intention queries [19], [20]. While effective for mode coverage, these designs often increase model complexity and memory usage, limiting scalability. Generative planners improve expressiveness: autoregressive Transformer approaches formulate motion generation as next-token prediction [21], [22], such as SMART and MotionLM [23], [24]. However, discrete sequential decoding can accumulate errors and hurt temporal coherence under tight interaction. Diffusion-based planners instead model continuous trajectory distributions and naturally produce temporally smooth samples, showing strong results in planning and simulation [6], [25]–[28]. Despite improved behavioral diversity, existing diffusion-based methods often struggle to balance agent interaction modeling with scene-conditioned consistency, leading to scene-inconsistent joint trajectories [6]. Moreover, these methods are susceptible to closed-loop distribution shift and objective misalignment, limiting robustness and reliability in real-world deployment [8]. In contrast, our method is built around this gap: *we retain diffusion's multimodal expressiveness while explicitly coupling scene-conditioned pre-training with RL post-training to optimize both closed-loop safety and efficiency.*

B. Reinforcement Post-training for Driving Planners

Reinforcement learning (RL) can further improve pretrained driving planners by combining sampling-based exploration with reward-driven policy optimization [8], [29], [30]. Recent RL-based planners mainly fine-tune two classes of generative models: (*i*) *autoregressive*, which can improve planning quality after fine-tuning (e.g., Plan-R1 [31], SMART-tiny-CLSFT [32], Think2Drive [33]), but often suffer from sequence–token objective mismatch and temporal instability induced by sequential decoding; and (*ii*) *diffusion*, which provide a temporally consistent decision process and diverse action sampling via probabilistic denoising in continuous space (e.g., TrajHF [13] and ReCogDrive [14]), making them particularly well-suited to RL's exploration and exploitation paradigm. However, existing approaches are typically *offline* and operate with limited reactive interaction, so limited rollouts often yield only marginal closed-loop gains [11], [13], [14]. Online RL remains difficult in *multi-agent* closed loop due to non-stationary interactions, compounding rollout errors, and high-variance gradient estimates. *We therefore aim to develop a stable online RL post-training framework by jointly designing (i) dense, well-shaped rewards and (ii) variance-gated updates, which together stabilize training and improve both closed-loop planning performance and cooperative behaviors.*

III. PROBLEM STATEMENT

We consider multi-agent planning in interactive traffic scenes. A scene is represented as $\mathcal{S} = (\mathbf{x}, \mathbf{u}, \mathbf{c})$, where $\mathbf{x} \in \mathbb{R}^{N_a \times T \times D_x}$ and $\mathbf{u} \in \mathbb{R}^{N_a \times T \times D_u}$ are state and control trajectories of N_a agents over horizon T, and $\mathbf{c} = (\mathbf{c}_a, \mathbf{c}_{mp}, \mathbf{c}_{tl})$ is scene context including: (i) *agent history* $\mathbf{c}_a$; (ii) *lane graph* $\mathbf{c}_{mp}$; and (iii) *traffic light* $\mathbf{c}_{tl}$. D_x, and D_u the dimensional features. We aim to learn a scene-conditioned *joint* planning policy $\pi_\theta(\cdot \mid \mathbf{c})$ that produces interaction-realistic multi-vehicle behaviors while improving closed-loop *safety* and *efficiency*.

We first pre-train a scene-conditioned multi-agent diffusion policy via imitation learning to capture multimodal interactions and generate joint future trajectories conditioned on $\mathbf{c}$. However, imitation-only diffusion policies can suffer from closed-loop distribution shift and objective misalignment. A key question is: *how can we preserve realistic interaction while promoting safety- and efficiency-oriented cooperative driving?*

We therefore propose MULTI-ORFT, a cooperative, scene-consistent multi-agent planner that post-trains the diffusion policy with stable online RL to improve closed-loop planning performance and strengthen cooperative driving behavior. Specifically, we sample rollouts $\mathbf{x}_0 \sim \pi_\theta(\mathbf{x}_0 \mid \mathbf{c})$ and optimize π_θ by maximizing the expected cumulative reward $\mathcal{J}(\pi_\theta)$.

IV. MULTI-AGENT DIFFUSION MODEL PRETRAINING

This section presents the implementation details of the scene condition-enhanced multi-agent trajectory diffusion model. Our design is motivated by the need to better balance inter-agent interaction modeling with scene conditioning.

A. Model Structure of the Multi-Agent Diffusion Planner

As shown in Fig. 2, the planner consists of a symmetric scene encoder $\mathcal{E}$ and a denoising decoder $\mathcal{D}_\theta$. The encoder maps raw context $\mathbf{c}$ to latent tokens $\tilde{\mathbf{c}} = \mathcal{E}(\mathbf{c})$ using query-centric attention in local frames, which capture pairwise relations [19]. The decoder $\mathcal{D} : (\tilde{\mathbf{c}}, \mathbf{u}_k, k) \mapsto \mathbf{u}_0$ is explicitly designed to strengthen scene consistency and constraint adherence of multi-agent trajectories through inter-agent self-attention and dual-path scene conditioning. It couples cross-attention for direct conditioning injection with AdaLN-Zero for flexible, stable conditional modulation and constraint enforcement throughout the denoising stack.

B. Symmetric Scene Context Modeling Encoder

The current scene inputs consist of *agent history* $\mathbf{c}_a \in \mathbb{R}^{N_a \times T_h \times D_a}$, *lane graph* $\mathbf{c}_{mp} \in \mathbb{R}^{N_m \times M_w \times D_p}$, and *traffic light* $\mathbf{c}_{tl} \in \mathbb{R}^{N_t \times D_t}$. Here N_a, N_m, and N_t denote the number of agents, polylines and traffic lights; T_h the number of historical steps; M_l the number of waypoints per polyline; and D_a, D_p, and D_t the corresponding feature sizes. They are initially tokenized into agent-history, map-polyline, and traffic-light tokens, then concatenated to fuse the features of the scene elements via a query-centric Transformer in local coordinates [19]. This design symmetrically aggregates relations among heterogeneous scene elements, yielding a unified context embedding $\tilde{\mathbf{c}} \in \mathbb{R}^{(N_a+N_m+N_t)\times D}$ for downstream denoising.

C. Multi-Agent Denoising Decoder

We diffuse joint control chunks in the action space of all vehicles and learn a denoiser $\mathcal{D}_\theta$ conditioned on $\tilde{\mathbf{c}}$ and diffusion step k. We embed noisy trajectories by rolling out controls through dynamics $f(\cdot)$ to obtain states $\mathbf{x}$, which are then encoded with an MLP. This keeps kinematic structure explicit during denoising. The denoiser then applies Transformer blocks with inter-agent self-attention.

Scene conditioning via cross-attention and AdaLN-Zero. Cross-attention alone can underutilize scene context under strong interaction coupling, leading to constraint violations [6]. We therefore couple (i) *cross-attention*, which injects $\tilde{\mathbf{c}}$ as keys/values to update trajectory tokens, with (ii) *AdaLN-Zero*, which provides stable layer-wise modulation conditioned on the diffusion step and dense road features. Specifically, since road-graph embedding $\tilde{\mathbf{c}}_{mp}$ can be sparse, we first densify it using a lightweight MLP-Mixer block that iteratively mixes information across the token and feature dimensions.

$$\begin{aligned} \tilde{\mathbf{c}}_{mp} &\leftarrow \tilde{\mathbf{c}}_{mp} + \mathrm{MLP}_{\text{tok}}\big(\tilde{\mathbf{c}}_{mp}^\top\big)^\top, \\ \tilde{\mathbf{c}}_{mp} &\leftarrow \tilde{\mathbf{c}}_{mp} + \mathrm{MLP}_{\text{ch}}(\tilde{\mathbf{c}}_{mp}). \end{aligned} \tag{1}$$

AdaLN then modulates each trajectory-token feature $\mathbf{x}$ by

$$\mathrm{AdaLN}(\mathbf{x}) = \big(1 + \gamma(k, \tilde{\mathbf{c}}_{mp})\big) \odot \mathbf{x} + \beta(k, \tilde{\mathbf{c}}_{mp}), \tag{2}$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are the scale and shift factors regressed from the sum of the diffusion-step embedding and the dense road features. We further regress a gating factor $\alpha(k, \tilde{\mathbf{c}}_{mp})$ applied to the residual branch. We initialize all α to zero so that the full decoder starts as an identity function and gradually learns to introduce conditioning, which strengthens conditional modulation and improves training stability.

Finally, the noise embedding is propagated to the final layer, which outputs the denoised action sequence $\mathbf{u}_0$. We train the decoder $\mathcal{D}_\theta(\mathbf{u}_k, k, \tilde{\mathbf{c}})$ by minimizing the following loss:

$$\mathcal{L}_{\mathcal{D}_\theta} = \mathbb{E}_{\mathbf{u}_0,\, k,\, \mathbf{u}_k \sim p_k(\cdot \mid \mathbf{u}_0)} \left[\mathcal{SL}_1(\mathbf{x}_0(\mathcal{D}_\theta(\mathbf{u}_k, k, \tilde{\mathbf{c}})) - \mathbf{x}_{gt}) \right], \tag{3}$$

where $\mathcal{SL}_1(\cdot)$ is the Smooth-ℓ_1 loss between the expert trajectory $\mathbf{x}_{gt}$ and the trajectory $\mathbf{x}_0$ induced by executing the denoised controls $\mathbf{u}_0$ through the dynamics $f(\cdot)$.

V. PLANNING-ORIENTED ONLINE REINFORCEMENT LEARNING

A multi-agent diffusion planner trained solely via imitation learning can capture general driving capabilities, but it often fails under closed-loop execution due to distribution shift and the scarcity of safety-critical interaction data. Furthermore, it fails to explicitly promote safety- and efficiency-oriented cooperative behaviors, relying solely on the BC loss.

To overcome these limitations, we propose a stable *online* RL post-training framework (Fig. 1) that enhances both robustness and cooperation in closed-loop planning. To support online optimization, we first formulate a two-level MDP. We then jointly design dense, well-shaped rewards and the VG-GRPO strategy to address instability during closed-loop training. Finally, we present the overall RL post-training pipeline.

A. Two-Level MDP

We now tailor the MDP to diffusion-policy optimization. Unlike standard policies that sample a_t in one shot, a diffusion policy produces the executed action u_t^0 through a *K-step stochastic denoising chain*. Treating the entire denoising chain as a black-box sampler in a single-level MDP leads to poor credit assignment across denoising steps and high-variance gradients, which is exacerbated in *multi-vehicle online* rollouts due to severe non-stationarity from coupled agent interactions. We therefore model denoising as an inner MDP $\mathcal{M}_{\text{DP}}$ with *analytically tractable Gaussian likelihoods* at each step, coupled with an outer environment-interaction MDP $\mathcal{M}_{\text{ENV}}$ that provides closed-loop rewards and constraints. Following [34], their composition yields the two-level MDP in Fig. 3. We use subscript t for the outer environment step and superscript k for the inner denoising step.

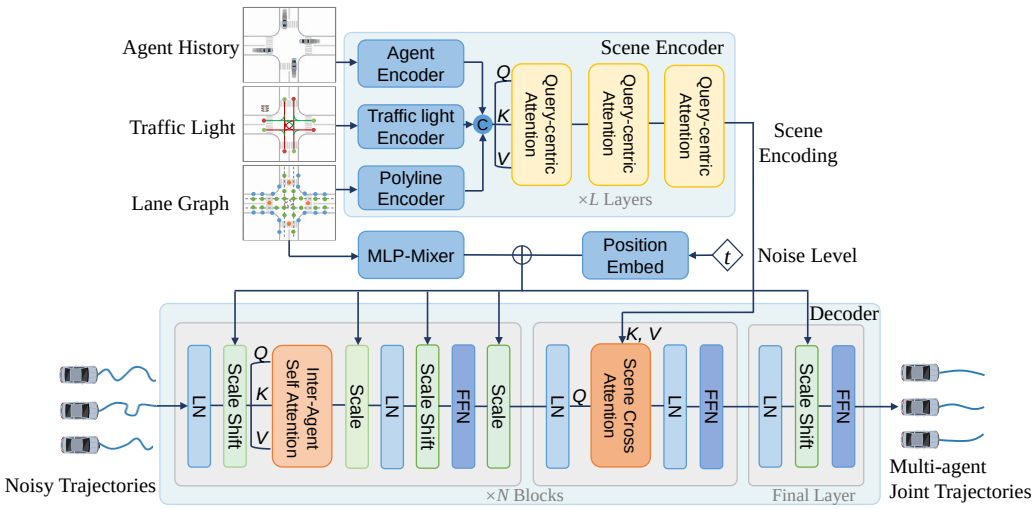


Fig. 2. Architecture of the multi-agent diffusion planner. A symmetric scene encoder captures local relational context with query-centric self-attention, and a denoising decoder generates joint multi-agent plans via inter-agent attention and scene-conditioned generation, augmented by AdaLN-Zero to provide flexible, stable conditional modulation and constraint enforcement throughout the denoising stack.

- **State.** $\bar{\mathbf{s}}_t^k = (\mathbf{c}_t, \mathbf{u}_t^k)$, where $\mathbf{c}_t$ is the scene observation at outer time step t and $\mathbf{u}_t^k$ is the intermediate denoising output at denoising step k.
- **Action.** $\bar{\mathbf{a}}_t^k = \mathbf{u}_t^k$, sampled from the step-wise reverse transition $\pi_\theta(\mathbf{u}_t^{k-1} \mid \mathbf{u}_t^k, \mathbf{c}_t)$; the final denoised action is executed in the outer MDP.
- **Initial distribution.** $\bar{\mathbf{s}}_0^K = (\mathbf{c}_0, \mathbf{u}_0^K) \sim \bar{\mathbf{P}}_0$, where $\mathbf{c}_0$ is drawn from the outer MDP's initial distribution and $\mathbf{u}_0^K \sim \mathcal{N}(0, \mathbf{I})$.
- **Transition.** The kernel $\bar{\mathbf{P}}(\bar{\mathbf{s}}_{t'}^{k'} \mid \bar{\mathbf{s}}_t^k, \bar{\mathbf{a}}_t^k)$ is defined as

$$\bar{\mathbf{s}}_{t'}^{k'} = \begin{cases} (\mathbf{c}_t, \bar{\mathbf{u}}_t^{k-1}) & (t', k') = (t, k-1), \ k > 0, \\ (\mathbf{c}_{t+1}, \mathbf{u}_{t+1}^K) & (t', k') = (t+1, K), \ k = 0, \end{cases} \tag{4}$$

  For $k > 0$, transitions occur within the inner denoising chain: $\mathbf{c}_t$ is fixed and the action is updated to $\mathbf{u}_t^{k-1}$. For $k = 0$ we execute $\mathbf{u}_t^0$ in the environment to obtain $\mathbf{c}_{t+1}$ and re-initialize $\mathbf{u}_{t+1}^K \sim \mathcal{N}(0, \mathbf{I})$.
- **Reward.** $\bar{\mathbf{R}}(\bar{\mathbf{s}}_t^k, \bar{\mathbf{a}}_t^k)$ are assigned only when denoising reaches $k = 0$:

$$\bar{\mathbf{R}}(\bar{\mathbf{s}}_t^k, \bar{\mathbf{a}}_t^k) = \begin{cases} 0 & k > 0, \\ R_{\text{ENV}}(\mathbf{c}_t, \mathbf{u}_t^0) & k = 0, \end{cases} \tag{5}$$

  where R_{ENV} is specified in Sec. V-B.
- **Objective.** The objective of the two-layer MDP is

$$\bar{\mathcal{J}}(\pi_\theta) = \mathbb{E}_{\pi_\theta, \bar{\mathbf{P}}, \bar{\mathbf{P}}_0} \left[\sum_{t \geq 0} \sum_{\tau \geq t} \gamma(\tau) \, \bar{R}\left(\bar{\mathbf{s}}_\tau^k, \bar{\mathbf{a}}_\tau^k\right) \right]. \tag{6}$$

- **Step-wise likelihood.** $\pi_\theta(\bar{\mathbf{a}}_t^k \mid \bar{\mathbf{s}}_t^k)$ can be evaluated by computing the diffusion-policy likelihood of each denoising step along the sampled denoising chain.

$$\begin{aligned} \pi_\theta(\bar{\mathbf{a}}_t^k \mid \bar{\mathbf{s}}_t^k) &= \pi_\theta(\mathbf{u}_t^{k-1} \mid \mathbf{u}_t^k, \mathbf{c}_t) \\ &= \mathcal{N}\left(\mathbf{u}_t^{k-1}; \mu\left(\mathbf{u}_t^k, \mathcal{D}_\theta\left(\mathbf{u}_t^k, k, \mathbf{c}_t\right)\right), \sigma_k^2 \mathrm{I}\right) \end{aligned} \tag{7}$$

  where $\pi_\theta(\mathbf{u}_t^{k-1} \mid \mathbf{u}_t^k, \mathbf{c}_t)$ is a Gaussian transition with mean $\mu(\cdot)$ calculated from $\mathbf{u}_t^k$ and denoiser output $\mathcal{D}_\theta\left(\mathbf{u}_t^k, k, \mathbf{c}_t\right)$, and variance σ_k^2 specified by the fixed noise schedule. Hence, the step-wise log-likelihood is analytically tractable.

B. Reward Modeling

Rewards determine both the optimization direction and the driving preference. We use simple but fine-grained rule-based rewards that generalize across diverse large-scale scenarios and follow a safety-first principle, with explicit incentives for efficiency. The overall reward is defined as

$$R_{\text{ENV}} = \sum_j w_c \, R_{\text{coll}}(j) + w_o \, R_{\text{offroad}}(j) + w_e \, R_{\text{eff}}(j). \tag{8}$$

where j is the step index within the reward horizon.

Collision. We define the collision reward through SAT-based oriented-box overlap checking, i.e., coll_overlaps$(\cdot)$:

$$R_{\text{coll}}(j) = \mathbb{1}[\text{coll_overlaps}(j)]. \tag{9}$$

Off-road. We define the off-road reward through signed distances of the four vehicle-box corners to lane boundaries, i.e., off_road$(\cdot)$:

$$R_{\text{offroad}}(j) = \mathbb{1}[\text{off_road}(j)]. \tag{10}$$

Efficiency. We define the efficiency reward by measuring normalized progress along a road centerline:

$$R_{\text{eff}}(j) = \max\left(\frac{s_{j+1} - s_j}{s_{\max}}, 0\right), \tag{11}$$

where s_j is the arc-length projection of the vehicle position onto the centerline and $s_{\max}$ normalizes the progress to $[0, 1]$. In multi-vehicle rollouts, we evaluate rewards at each step over the horizon and average them across the controlled vehicles. This formulation provides dense and well-shaped reward evaluations that enable finer-grained discrimination

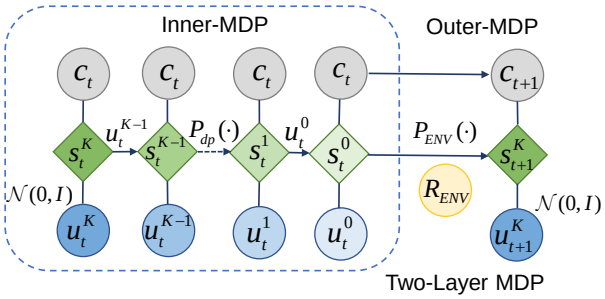


Fig. 3. Two-level MDP for online RL. An inner denoising MDP is coupled with an outer environment-interaction MDP, and denoising step-wise Gaussian likelihoods make denoising process can be optimized via policy gradients.

among sampled trajectories, thus yielding stable learning signals. To avoid *trajectory* dynamic-quality collapse when emphasizing safety, we use three built-in safeguards: dynamics-aware denoising with rollout through $f(\cdot)$ to ensure trajectory feasibility, KL anchoring to the pre-training prior, and short-horizon execute-then-replan in closed loop.

C. Variance-Gated Group Relative Policy Optimization

With the two-level MDP (Sec. V-A) and reward (Sec. V-B) in place, we optimize θ to maximize the objective $\bar{\mathcal{J}}(\pi_\theta)$, where π_θ is the pretrained diffusion policy induced by the reverse denoising chain. We propose VG-GRPO for online refinement of multi-agent diffusion policy. Our design is built on the critic-free GRPO paradigm [35] and extends it with variance-gated advantages and denoising-aware optimization to stabilize training under multi-agent non-stationarity. This is a functional coupling rather than a loose combination: diffusion provides grouped multimodal candidates each step, and VG-GRPO directly consumes their relative ranking signal for online updates.

At each outer environment step t, we fix the encoded context $\tilde{\mathbf{c}}_t$ and sample a group of G candidate rollouts from the old policy $\pi_{\theta_{\text{old}}}$. Let $i \in \{1, \ldots, G\}$ index samples in the group. Each sample yields a scalar reward $r_{t,i} := \bar{\mathbf{R}}_{t,i}$, forming the within-group reward set $\{r_{t,1}, \ldots, r_{t,G}\}$.

VG-GRPO objective. We maximize the following objective

$$\begin{aligned}\bar{\mathcal{J}}(\theta) = \mathbb{E}_{\left(\{\bar{\mathbf{a}}_{t,i}^k\}_{i=1}^G, \bar{\mathbf{s}}_t\right) \sim \pi_{\theta_{old}}} \frac{1}{G} \sum_{i=1}^{G} -\beta \mathbb{D}_{KL}\left(\pi_\theta || \pi_{ref}\right) \\ + \left(\min\left(\rho_{t,i}^k, \text{clip}\left(\rho_{t,i}^k, 1-\varepsilon^-, 1+\varepsilon^+\right)\right) \gamma_{\text{denoise}}^k A_{t,i}\right),\end{aligned} \tag{12}$$

where $\rho_{t,i}^k$ is the importance ratio, computed from step-wise likelihoods under the two-level MDP in Sec. V-A.

$$\rho_{t,i}^k := \frac{\pi_\theta\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)}{\pi_{\theta_{old}}\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)}, \tag{13}$$

and $[\varepsilon^-, \varepsilon^+]$ is the DAPO-style clipping ratio that stabilizes training and encourages exploration, and $\gamma_{\text{denoise}}^k \in (0, 1]$ is a denoising-step discount factor that downweights the policy-gradient contributions of noisier steps, thereby improving training stability [36]. To prevent mode collapse during exploration and preserve general driving capabilities, we regularize post-training with a KL term to the reference policy (set to the pretrained policy here), using the non-negative unbiased estimator [37] and weighting it by β to control the regularization strength:

$$\mathbb{D}_{KL}\left(\pi_\theta || \pi_{ref}\right) = \frac{\pi_{ref}\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)}{\pi_\theta\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)} - \log \frac{\pi_{ref}\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)}{\pi_\theta\left(\bar{\mathbf{a}}_{t,i}^k \mid \bar{\mathbf{s}}_t^k\right)} - 1, \tag{14}$$

Variance-gated advantages. Standard within-group normalization can induce training instability. In simple cases (e.g., short rollouts or near-stationary vehicles), sampled rewards become nearly identical; after GRPO's mean–std normalization, the resulting advantages collapse toward zero, yielding vanishing policy gradients. This amplifies the sensitivity of minibatch gradients to noise and directly destabilizes training. To address this issue, we introduce a variance gated mechanism, inspired by [38] and adapted to fine-grained selection over group-based trajectory rollouts. The group relative advantage for sample i is defined as

$$A_{t,i} = \begin{cases} \text{drop this group}, & \text{std} \le \text{std}_1, \\ r_{t,i} - \text{mean}(\{r_{t,1}, \ldots, r_{t,G}\}), & \text{std}_1 < \text{std} \le \text{std}_2, \\ \dfrac{r_{t,i} - \text{mean}(\{r_{t,1}, \ldots, r_{t,G}\})}{\text{std}(\{r_1, \ldots, r_G\})}, & \text{std} > \text{std}_2. \end{cases} \tag{15}$$

During policy sampling, we measure group diversity via the within-group standard deviation of rewards. (i) If the within-group standard deviation is near zero, rollouts are essentially identical, so we discard the group to avoid vanishing gradients. (ii) If it is small but nonzero, the differences may be noise-driven. Normalization would amplify this noise and corrupt the update direction, so we keep the raw differences to retain sample utility while reducing instability. (iii) If it is large, rollouts exhibit meaningful quality gaps and provide informative gradients; we then apply standard group-relative normalization to compute advantages. This variance gate retains informative samples and reduces gradient variance, thereby stabilizing typical GRPO for online RL training of multi-agent diffusion policies.

D. Reinforcement Learning Post-training Framework

We implement the online RL framework in Fig. 1 with three stages: policy rollout, reward evaluation, and policy update. At each outer step t, The old policy $\pi_{\theta_{\text{old}}}$ samples a group of G candidate rollouts $\bar{\boldsymbol{a}}_{t,i}^k$ from the latest context $\bar{\boldsymbol{s}}_t^k$, scores each rollout by rule-based safety and efficiency rewards $\bar{\mathbf{R}}_{t,i}$, and updates the multi-agent planner with **VG-GRPO** (Sec. V-C). We use a rolling horizon (plan T_p, execute first T_a), execute the best-performing sample for environment interaction, and retain *all* group samples for relative-advantage computation.

We store grouped tuples in the rollout buffer $\mathcal{D}_{\text{itr}}$:

$$\left(\{\bar{\mathbf{s}}_t^k\},\ \{\bar{\mathbf{a}}_{t,i}^k\}_{i=1}^G,\ \{\bar{\mathbf{R}}_{t,i}\}_{i=1}^G,\ k\right), \tag{16}$$

During updates, we sample mini-batches from $\mathcal{D}_{\text{itr}}$, compute variance-gated group relative advantages (Eq. (15)), and optimize the planner parameters by maximizing the VG-GRPO objective (Eq. (12)).

VI. Simulation Results and Discussion

This section reports closed-loop evaluation of the proposed planner. We first describe datasets, metrics, and implementation details, then present benchmark comparisons and ablation studies. The evaluation verifies three claims: closed-loop safety and efficiency gains; robustness of the online post-training design choices; and the contribution of the AdaLN-Zero conditional module and VG-GRPO.

A. Setup

Dataset. For pre-training, we use the Waymo Open Motion Dataset (WOMD) [17], containing 486,995 training scenarios and 44,097 validation scenarios; each scenario covers 9 s of real traffic with all participant trajectories and map topology. We perform closed-loop evaluation and ablation studies across 41,590 Testing Interactive scenarios. For post-training, we uniformly sample 20,756 scenarios from the WOMD Validation Interactive split.

Implementation Details. Scene inputs include up to $N_v = 32$ agents, $N_m = 256$ polylines with $M_w = 30$ waypoints each, and $N_t = 16$ traffic lights. The planner generates a future control sequence of length $T_p = 80$ with a 0.1 s time step. We discard the entire history and condition on the current state. In closed-loop testing, we execute $T_a = 10$ steps.

The scene encoder uses $L = 6$ query-centric Transformer layers with hidden size $D = 256$. The denoising decoder alternates two block types for three rounds (6 Transformer layers total). MLP-Mixer token/channel dimensions are 64/128. Pre-training uses the log noise schedule in [6] with $\bar{\alpha}_{\min} = 10^{-9}$, scaling 0.0031, and $K = 20$. We pre-train with AdamW (weight decay 0.01), learning rate 2×10^{-4} (3000-step warmup; $\times 0.02$ decay every 3000 steps), gradient clipping 1.0, BF16 on 4×RTX 4090 for 30 epochs (global batch size 32).

Post-training uses group size $G = 10$. Each rollout batch is optimized for 1 epoch with mini-batch size 16 for 10M fine-tuning steps. Following [8], we use 4 s reward horizon, reward weights (collision/off-road/progress) $8/1/4$, KL weight $\beta = 0.1$, and learning rate 10^{-5}. For effective exploration, we use DAPO-style clipping $[-0.15, 0.2]$ [38], and clamp a minimum sampling-time standard deviation of $\sigma_{\text{min}}^{\text{sam}} = 0.2$. For training stability, we clamp the per-step log-likelihood standard deviation to $\sigma_{\text{min}}^{\text{prob}} = 0.1$. We set gated thresholds to $\text{std}_1/\text{std}_2 = 0.03/0.06$ (10%/20% of the reward standard deviation based on), guided by signal-to-noise [39] and advantage-collapse analysis [40] and use $\gamma_{\text{denoise}} = 0.9$. Post-training runs in BF16 on one NVIDIA 5090 GPU.

Metrics. We report collision rate (CR), off-road rate (OR), average speed (AS), average displacement error (ADE), and kinematic infeasibility (Kin). We treat CR, OR, and AS as *primary* closed-loop objectives reflecting safety and traffic efficiency, and use ADE and Kin as *secondary* diagnostics for imitation fidelity and physical feasibility. CR checks SAT-based oriented-box overlap; OR checks drivable-area boundary crossing; AS is mean per-step displacement; ADE is the ℓ_2 position error to ground truth; and Kin counts violations of acceleration and curvature bounds with limits $6\,\text{m/s}^2$ and $0.3\,\text{m}^{-1}$. For fairness, all methods are evaluated with the same simulator configuration, including scenario split, agent count, horizon, and replanning frequency; values with $\pm$ report mean and standard deviation over repeated closed-loop evaluations in the main comparison tables, while ablation tables report point estimates under the same protocol for compactness.

TABLE I
Closed-loop benchmark on the WOMD testing interactive split (primary: CR, OR, AS; secondary: ADE, Kin)

Method	Primary closed-loop objectives			Secondary diagnostics	
	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
TrafficBotsV1.5 [41]	2.74±0.21	1.79±0.14	8.03±0.48	1.68±0.09	0.26±0.02
SMART-large [23]	2.22±0.09	1.58±0.10	8.34±0.30	1.30±0.01	**0.21±0.01**
VBD [6]	2.46±0.14	1.92±0.18	8.08±0.52	1.41±0.02	0.24±0.01
SMART-tiny-CLSFT [32]	2.10±0.10	1.53±0.12	8.47±0.44	**1.23±0.03**	0.25±0.02
Multi-ORFT	**1.89±0.12**	**1.36±0.08**	**8.61±0.46**	1.36±0.04	0.32±0.03

B. Evaluation

This subsection evaluates interactive trajectory generation using both benchmark tables and representative qualitative cases.

WOMD Results. We compare four strong open-source baselines: VBD [6], SMART-large [23], SMART-tiny-CLSFT [32], and TrafficBotsV1.5 [41]. They cover autoregressive, behavior-cloning, and diffusion paradigms. Each scenario runs for 8 s in closed loop with replanning at 1 Hz. Table I shows that Multi-ORFT leads on all three primary closed-loop objectives. Compared with the strong SMART-tiny-CLSFT baseline, which uses targeted closed-loop supervised fine-tuning, our method reduces CR from 2.10 to 1.89 (-10.0%) and OR from 1.53 to 1.36 (-11.1%), while increasing AS from 8.47 to 8.61 (+1.66%). Relative to the pretrained policy in Table II, online post-training yields 7.4% lower CR, 19.0% lower OR, and 3.0% higher AS. This is consistent with the mechanism: VG-GRPO directly optimizes safety and efficiency rewards in closed loop, enabling better interaction outcomes under distribution shift.

Post-training Methods. We further compare four post-training strategies, SFT, DPO, offline RL, and online RL, as shown in Table II. All methods improve over the pre-training-only policy in part, but with different objective alignment. SFT mainly improves imitation-oriented diagnostics (ADE). DPO and offline RL improve selected indicators but can degrade the balance between safety and efficiency. Multi-ORFT with online RL provides the most consistent co-improvement on the primary closed-loop objectives CR, OR, and AS, reflecting the benefit of coupling diffusion group sampling with VG-GRPO.

Online RL shows a mild ADE increase, indicating a distributional trade-off: closed-loop exploration can reduce one-step imitation accuracy while improving interaction robustness. Kin remains low across methods, all below 0.4%; compared with the pretrained policy, it increases by at most 0.07 percentage points from 0.25 to 0.32, indicating no dynamic-quality collapse under safety-oriented optimization.

Qualitative Visualization. We then compare pre-training and RL post-training in an out-of-distribution interaction case. The pre-trained policy fails to resolve a left-turn conflict between

TABLE II
CLOSED-LOOP COMPARISON OF POST-TRAINING METHODS (PRIMARY: CR, OR, AS; SECONDARY: ADE, KIN)

Method	Primary closed-loop objectives			Secondary diagnostics	
	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
Pre-trained only	2.04±0.11	1.68±0.10	8.36±0.42	1.28±0.02	**0.25±0.02**
SFT	2.01±0.07	1.64±0.06	8.37±0.36	**1.15±0.015**	**0.25±0.01**
DPO	1.97±0.13	1.58±0.09	8.15±0.39	1.33±0.04	0.27±0.01
Offline RL	2.18±0.09	1.82±0.14	**8.98±0.68**	1.37±0.05	0.26±0.02
MULTI-ORFT (Online RL)	**1.89±0.12**	**1.36±0.08**	8.61±0.46	1.36±0.04	0.32±0.03

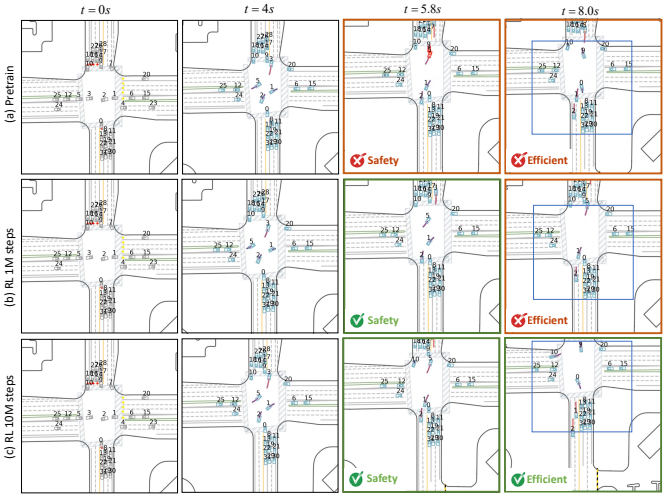


Fig. 4. Qualitative comparison between pre-training and post-training on an 8-second closed-loop rollout from the ego view of Vehicle 1. One-second history trajectories are shown for context.

Vehicle 5 and Vehicle 9, resulting in collision at $t = 5.8$ s. After 1M RL steps, the policy learns conservative yielding behavior, which improves safety but reduces efficiency. After 10M steps, the policy maintains safety while recovering decisiveness, yielding higher overall traffic progress. This progression shows how reward-driven post-training reshapes interaction strategy.

C. Ablations

Unless otherwise stated, ablations use the same training pipeline and evaluation protocol as Section VI.

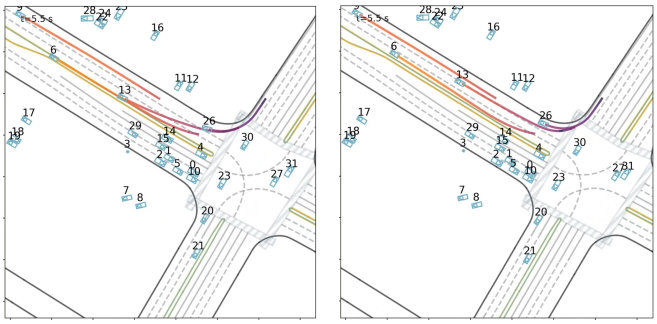
Impact of AdaLN-Zero Module. AdaLN-Zero improves conditional generation quality. Table III shows gains on multiple metrics, especially off-road rate. The mechanism is direct: condition-driven modulation reshapes denoising features and strengthens road-geometry awareness, which is harder to achieve with cross-attention alone. Better boundary compliance in generation then transfers to improved closed-loop stability.

Fig. 5 provides a qualitative comparison on sharp right-turn cases. Without AdaLN-Zero, Vehicles 6 and 13 drift outward and hit the median boundary. With AdaLN-Zero, turn trajectories are more compact and maintain safer boundary margins.

Impact of the Variance-gated Mechanism. Table IV analyzes the effect of our *variance-gated* mechanism via ablations over several settings, including a non-gated baseline and multiple threshold combinations. Without variance gating, post-training collapses at ∼0.5M steps, evidenced by persistently large gradient-norm fluctuations and a sudden spike in the KL term, evaluating the last stable checkpoint also shows degraded performance across metrics. We then vary the gating thresholds. With $\text{std}_1/\text{std}_2 = 0.00/0.06$, collapse is delayed to ∼2.0M steps and key closed-loop safety and efficiency metrics improve slightly, indicating that std_2 contributes to training stability. In contrast, $0.03/0.06$ and $0.03/0.09$ prevent collapse completely, suggesting that std_{low} is critical for stability by mitigating advantage degeneration when within-group samples are nearly identical. Among stable settings, $0.03/0.06$ achieves the best performance, a larger std_2 appears overly conservative and weakens the effective learning signal, suggesting that moderate thresholds strike a better balance between stability and performance gains.

TABLE III
ABLATION ON ADALN-ZERO MODULE

Setting	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
w/o AdaLN-Zero	2.11	2.05	**8.40**	1.30	0.26
with AdaLN-Zero	**2.04**	**1.68**	8.36	1.28	0.25



(a) without AdaLN-Zero module (b) with AdaLN-Zero module

Fig. 5. Qualitative comparison with and without the AdaLN-Zero module

TABLE IV
ABLATION OF THE VARIANCE-GATED MECHANISM (PERFORMANCE IS EVALUATED AT THE LAST CHECKPOINT BEFORE COLLAPSE)

Gating $\text{std}_1/\text{std}_2$	Collapse step	CR[%]↓	OR[%]↓	AS$[m/s]$↑	ADE$[m]$↓	Kin[%]↓
w/o gating	≈ 0.5M	2.15	2.03	8.05	1.74	0.40
0.03/0.06	–	**1.89**	**1.36**	**8.61**	1.36	0.32
0.00/0.06	≈ 2.0M	2.01	1.57	8.42	1.42	0.30
0.03/0.09	–	1.96	1.50	8.49	**1.30**	**0.29**

VII. CONCLUSION

In this paper, we introduced MULTI-ORFT, a cooperative multi-agent planner that couples condition-enhanced diffusion pre-training with stable online RL post-training in reactive closed-loop environments. For pre-training, MULTI-ORFT strengthens scene consistency and road adherence by combining inter-agent self-attention with dual-path scene conditioning. For post-training, MULTI-ORFT achieves stable online learning by combining dense, well-shaped rewards with our proposed VG-GRPO, mitigating advantage collapse and gradient instability while strengthening closed-loop cooperative behaviors. Extensive experiments demonstrate superior closed-loop planning

performance on key *safety* and *efficiency* metrics, outperforming state-of-the-art baselines on WOMD and across alternative post-training paradigms.

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